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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CAPSTONE PROJECT

SMART WATER DISTRIBUTION MONITORING USING GRAPH-BASED NETWORK ANALYSIS

CSA0309

UNDER THE SUPERVISION OF:
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Introduction

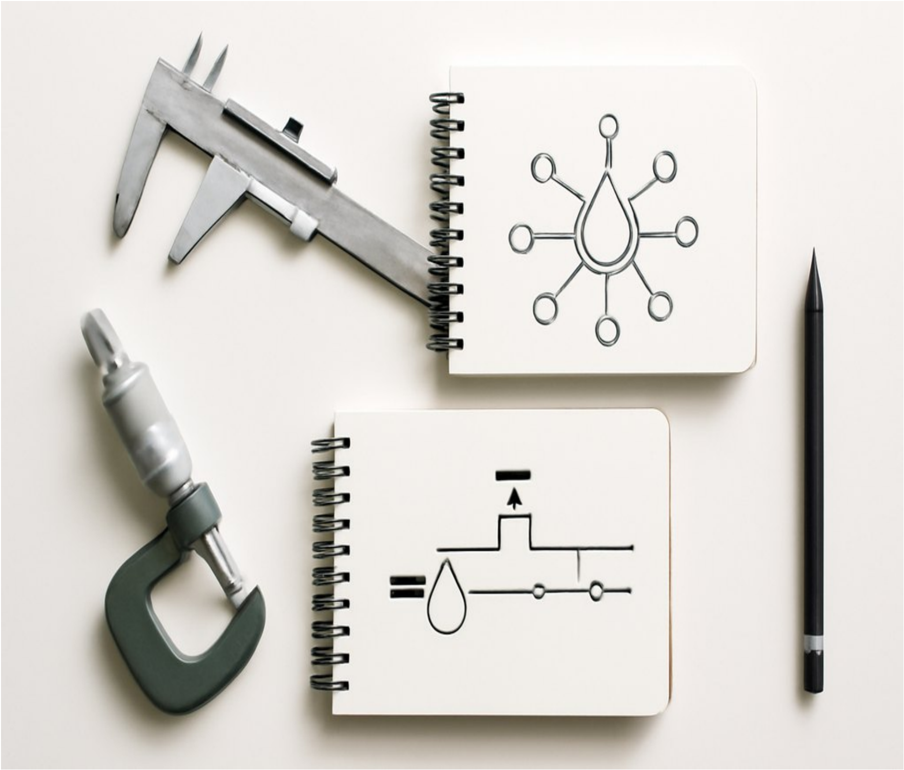
- Smart Water Distribution Monitoring is an intelligent approach for managing modern water supply networks.
- It uses graph-based network analysis to study the structure and flow of the distribution system.
- The system monitors important points such as water sources, storage tanks, pipelines, and distribution junctions.
- It helps identify network failures, leakages, and inefficient water flow.
- The main goal is to provide efficient water distribution, reduce wastage, and ensure continuous water supply.

SMART WATER DISTRIBUTION OVERVIEW

- **Real-time Monitoring:** Uses IoT sensors to continuously monitor water flow, pressure, tank levels, and water quality.
- **AI-Based Optimization:** Employs Artificial Intelligence and Machine Learning to predict water demand and optimize water distribution.
- **Leak Detection:** Automatically identifies leaks, pipe bursts, and abnormal water usage to minimize water loss.
- **Smart Automation:** Controls pumps and valves automatically for efficient and uninterrupted water supply.
- **Sustainable Water Management:** Reduces water wastage, lowers operational costs, and supports smart city infrastructure.

OBJECTIVES

- **To develop a smart water distribution monitoring system** that collects real-time data from sensors installed across the water network.
- **To model the water distribution network as a graph**, where nodes represent reservoirs, tanks, and junctions, and edges represent pipelines.
- **To analyze the network using graph algorithms** for efficient water flow, shortest path identification, and optimal resource distribution.
- **To detect leaks, blockages, and abnormal water flow** by identifying anomalies in the network graph and sensor data.
- **To improve the efficiency, reliability, and sustainability** of water distribution through real-time monitoring, predictive analysis, and da driven decision-making.



PROBLEM STATEMENT

Traditional water distribution systems have limited real-time monitoring.
Leakages and blockages may go unnoticed, causing significant water wastage.
Uneven water flow can lead to low pressure and irregular water supply.
It is difficult to identify the critical points and weak connections in large networks.
A smart graph-based system is needed to analyze the network, detect problems, and improve water distribution efficiency.

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LITERATURE REVIEW

Author & Year	Title	Methodology	Key Findings	Limitation
Smith et al. (2022)	IoT-Based Smart Water Distribution System	Implemented IoT sensors to monitor water flow, pressure, and tank levels in real time.	Achieved real-time data collection and remote monitoring, improving operational efficiency.	Limited fault prediction capability and not suitable for large-scale networks.
Kumar & Singh (2023)	Leak Detection in Water Networks Using Machine Learning	Applied ML algorithms (SVM, Random Forest) to detect leakages using historical data.	Improved leak detection accuracy and reduced water loss.	Requires a large amount of historical data and high computational resources.
Chen et al. (2021)	Graph-Based Analysis for Water Distribution Networks	Modeled the water distribution system as a graph and used graph algorithms for flow optimization.	Improved flow efficiency and identified critical nodes and bottlenecks in the network.	Did not support real-time monitoring and dynamic updates.
Ahmed et al. (2024)	AI-Enabled Smart Water Management System	Integrated IoT sensors with AI models for demand prediction and automated control of valves and pumps.	Achieved accurate demand prediction and efficient resource management, reducing water wastage.	High implementation cost and system complexity.

RESEARCH GAP

Existing systems focus on either IoT monitoring or AI-based prediction. There is a need for an integrated approach that combines **Graph-Based Network Analysis** with **real-time monitoring** to achieve efficient water flow optimization and early fault detection in water distribution networks.

Solution

- Represent the water distribution system as a graph of nodes and pipelines.

- Monitor water flow, pressure, and network conditions in real time.
- Use graph algorithms to identify critical paths, disconnected areas, and weak points.
- Detect possible leakages, blockages, and abnormal flow patterns.
- Provide data-driven decisions to reduce water wastage and improv distribution efficiency.

Module 1: Water Network Modeling

Objective: Create a digital representation of the water distribution network.

- Model **pipes, junctions, tanks, pumps, and reservoirs**
- Represent the network as interconnected components
- Simulate **water flow, pressure, and demand**
- Identify critical points and possible weak areas
- Provides the foundation for monitoring and analysis

Key Output: Digital water network model for simulation and analysis.

Module 2: Sensor Data Collection

Objective: Collect real-time information from the water network.

- Deploy sensors at important network locations
- Collect **pressure, flow rate, water level, and quality data**
- Continuously transmit sensor readings to the system
- Store historical data for further analysis
- Detect abnormal readings and missing data

Key Output: Reliable real-time and historical sensor dataset.

Module 3: Graph-Based Network Analysis

Objective: Analyze the water network using graph-based techniques.

- Represent **junctions as nodes** and **pipes as edges**
- Analyze network connectivity and structure
- Identify critical nodes and important pipelines
- Detect isolated or vulnerable network sections
- Study flow paths and network dependencies

Key Output: Graph-based representation showing network structure and critic

components.

Module 4: Leak Detection & Fault Identification

Objective: Detect leaks and identify abnormal network conditions.

- Compare expected and actual **flow/pressure values**
- Detect unusual patterns using sensor data
- Identify potential **leaks, pipe failures, and blockages**
- Locate the probable affected network section
- Generate alerts for quick maintenance action

Key Output: Early detection and identification of network faults.

Module 5: Smart Monitoring Dashboard

Objective: Provide a centralized interface for real-time network monitoring.

- Display live **flow, pressure, and sensor status**
- Show network conditions using interactive visualizations
- Highlight detected leaks and faults
- Provide alerts and warning notifications
- Support historical data and trend analysis

Key Output: User-friendly dashboard for real-time water network monitoring.

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Module 6: Optimization & Decision Support

Objective: Improve network efficiency and support operational decisions.

- Optimize **water flow and pressure distribution**
- Recommend suitable maintenance actions
- Reduce water losses and operational costs
- Prioritize critical pipes and repair locations
- Support data-driven planning and resource allocation

Key Output: Optimized network operation with intelligent decision support.

CONCLUSION

- Graph-based analysis provides an efficient way to monitor water distribution networks.
- It helps detect leakages, blockages, and abnormal water flow.
- The system improves water distribution and reduces wastage.
- Real-time monitoring supports quick identification of network problems.
- Overall, it provides a smart, reliable, and sustainable solution for water management.

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THANK YOU